

Lab Preparation: TP2 High-Density Network

WAN Link Simulation Between Two Routers

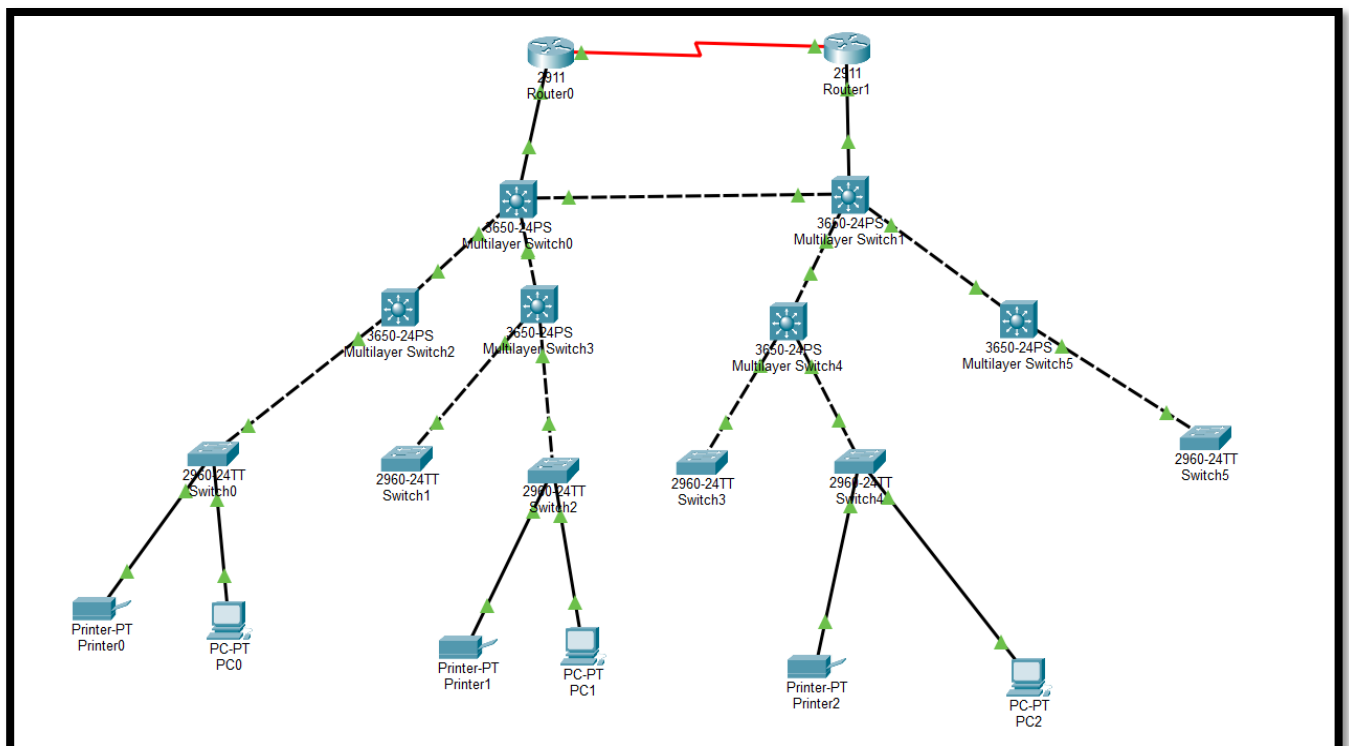
Network Simulation Lab

3-Tier Architecture with Catalyst 3650

Last Name : Belkacem

First Name : Nacim

GRP : 1.2



Complete configurations of R1 :

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! Router name
Router(config)#hostname R1
R1(config)#
R1(config)#! WAN interface configuration (DCE side)
R1(config)#interface Serial0/2/0
R1(config-if)# description WAN_LINK_TO_R2
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# clock rate 64000
This command applies only to DCE interfaces
R1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
R1(config-if)# exit
R1(config)#
R1(config)#! LAN interface configuration with sub-interfaces
R1(config)#interface GigabitEthernet0/0
R1(config-if)# description LAN_CONNECTION_TO_CORE_C1
R1(config-if)# no shutdown

R1(config-if)# exit
R1(config)#
R1(config)#! VLAN 10 sub-interface
R1(config)#interface GigabitEthernet0/0.10
R1(config-subif)# description VLAN10_DATA
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# no shutdown
R1(config-subif)# exit
R1(config)#
R1(config)#! VLAN 20 sub-interface
R1(config)#interface GigabitEthernet0/0.20
R1(config-subif)# description VLAN20_DATA
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# no shutdown
R1(config-subif)# exit
R1(config)#
R1(config)#! OSPF routing configuration
R1(config)#router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
R1(config-router)# network 192.168.20.0 0.0.0.255 area 0
R1(config-router)# exit
R1(config)#
R1(config)#! Console line configuration
R1(config)#line console 0
R1(config-line)# password cisco
R1(config-line)# login
```

```
R1(config-line)# exit
R1(config)#
R1(config)#! VTY lines configuration
R1(config)#line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)#
R1(config)#! Enable password
R1(config)#enable secret class
R1(config)#
R1(config)#! Save configuration
R1(config)#end
```

```
R1#write memory
```

```
Building configuration...
```

```
[OK]
```

Complete configurations of R2 :

```
R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#! Router name
R2(config)#hostname R2
R2(config)#
R2(config)#! WAN interface configuration (DTE side)
R2(config)#interface Serial0/3/0
R2(config-if)# description WAN_LINK_TO_R1
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# ! NO clock rate on DTE side
R2(config-if)# no shutdown

R2(config-if)# exit
R2(config)#
R2(config)#! LAN interface configuration with sub-interfaces
R2(config)#interface GigabitEthernet0/0
R2(config-if)# description LAN_CONNECTION_TO_CORE_C2
R2(config-if)# no shutdown

R2(config-if)# exit
R2(config)#
R2(config)#! VLAN 30 sub-interface
R2(config)#interface GigabitEthernet0/0.30
R2(config-subif)# description VLAN30_DATA
R2(config-subif)# encapsulation dot1Q 30
R2(config-subif)# ip address 192.168.30.1 255.255.255.0
R2(config-subif)# no shutdown
R2(config-subif)# exit
R2(config)#
```

```

R2(config)#! VLAN 40 sub-interface
R2(config)#interface GigabitEthernet0/0.40
R2(config-subif)# description VLAN40_DATA
R2(config-subif)# encapsulation dot1Q 40
R2(config-subif)# ip address 192.168.40.1 255.255.255.0
R2(config-subif)# no shutdown
R2(config-subif)# exit
R2(config)#! OSPF routing configuration
R2(config)#router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
R2(config-router)# network 192.168.30.0 0.0.0.255 area 0
R2(config-router)# network 192.168.40.0 0.0.0.255 area 0
R2(config-router)# exit
R2(config)#
R2(config)#! Console line configuration
R2(config)#line console 0
R2(config-line)# password cisco
R2(config-line)# login
R2(config-line)# exit
R2(config)#
R2(config)#! VTY lines configuration
R2(config)#line vty 0 4
R2(config-line)# password cisco
R2(config-line)# login
R2(config-line)# exit
R2(config)#
R2(config)#! Enable password
R2(config)#enable secret class
R2(config)#
R2(config)#! Save configuration
R2(config)#end
R2#write memory
Building configuration...
[OK]

```

Results of (show ip route) commands from R1 and R2 :

R1	R2
<pre> R1>show ip route Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is not set 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks C 10.0.0.0/30 is directly connected, Serial0/2/0 L 10.0.0.1/32 is directly connected, Serial0/2/0 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10 L 192.168.10.1/32 is directly connected, GigabitEthernet0/0.10 192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20 L 192.168.20.1/32 is directly connected, GigabitEthernet0/0.20 O 192.168.30.0/24 [110/65] via 10.0.0.2, 01:08:54, Serial0/2/0 O 192.168.40.0/24 [110/65] via 10.0.0.2, 01:08:54, Serial0/2/0 </pre>	<pre> R2>show ip route Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is not set 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks C 10.0.0.0/30 is directly connected, Serial0/3/0 L 10.0.0.2/32 is directly connected, Serial0/3/0 O 192.168.10.0/24 [110/65] via 10.0.0.1, 01:15:19, Serial0/3/0 O 192.168.20.0/24 [110/65] via 10.0.0.1, 01:15:09, Serial0/3/0 192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.30.0/24 is directly connected, GigabitEthernet0/0.30 L 192.168.30.1/32 is directly connected, GigabitEthernet0/0.30 192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.40.0/24 is directly connected, GigabitEthernet0/0.40 L 192.168.40.1/32 is directly connected, GigabitEthernet0/0.40 R2> </pre>

Results of show controllers serial from R1 and R2 :

R1	R2
<pre>R1>show controllers serial0/2/0 Interface Serial0/2/0 Hardware is PowerQUICC MPC860 DCE V.35, clock rate 2000000 idb at 0x81081AC4, driver data structure at 0x81084AC0 SCC Registers: General [GSMR]=0x2:0x00000000, Protocol-specific [PSMR]=0x8 Events [SCCE]=0x0000, Mask [SCCM]=0x0000, Status [SCCS]=0x00 Transmit on Demand [TODR]=0x0, Data Sync [DSR]=0x7E7E Interrupt Registers: Config [CICR]=0x00367F80, Pending [CIPR]=0x0000C000 Mask [CIMR]=0x00200000, In-srv [CISR]=0x00000000 Command register [CR]=0x580 Port A [PADIR]=0x1030, [PAPAR]=0xFFFF [PAODR]=0x0010, [PADAT]=0xCBFF Port B [PBDIR]=0x09C0F, [PBPAPAR]=0x0800E [PBODR]=0x00000, [PBDAT]=0x3FFFD Port C [PCDIR]=0x00C, [PCPAR]=0x200 [PCSO]=0xC20, [PCDAT]=0xDF2, [PCINT]=0x00F Receive Ring rmd(68012830): status 9000 length 60C address 3B6DAC4 rmd(68012838): status B000 length 60C address 3B6D444 Transmit Ring tmd(680128B0): status 0 length 0 address 0 tmd(680128B8): status 0 length 0 address 0 tmd(680128C0): status 0 length 0 address 0 tmd(680128C8): status 0 length 0 address 0 tmd(680128D0): status 0 length 0 address 0 tmd(680128D8): status 0 length 0 address 0 tmd(680128E0): status 0 length 0 address 0 tmd(680128E8): status 0 length 0 address 0 tmd(680128F0): status 0 length 0 address 0 tmd(680128F8): status 0 length 0 address 0 tmd(68012900): status 0 length 0 address 0 tmd(68012908): status 0 length 0 address 0 tmd(68012910): status 0 length 0 address 0 tmd(68012918): status 0 length 0 address 0 tmd(68012920): status 0 length 0 address 0 tmd(68012928): status 2000 length 0 address 0 tx_limited=1(2) SCC GENERAL PARAMETER RAM (at 0x68013C00) Rx BD Base [RBASE]=0x2830, Fn Code [RFCR]=0x18 Tx BD Base [TBASE]=0x28B0, Fn Code [TFCR]=0x18 Max Rx Buff Len [MRBLR]=1548 Rx State [RSTATE]=0x0, BD Ptr [RBPTR]=0x2830 Tx State [TSTATE]=0x4000, BD Ptr [TBPTR]=0x28B0 SCC HDLC PARAMETER RAM (at 0x68013C38) CRC Preset [C_PRES]=0xFFFF, Mask [C_MASK]=0xF0B8 Errors: CRC [CRCEC]=0, Aborts [ABTSC]=0, Discards [DISFC]=0 Nonmatch Addr Cntr [NMARC]=0 Retry Count [RETRC]=0 Max Frame Length [MFLR]=1608 Rx Int Threshold [RFTHR]=0, Frame Cnt [RFCNT]=0 User-defined Address 0000/0000/0000/0000 User-defined Address Mask 0x0000 buffer size 1524 PowerQUICC SCC specific errors: 0 input aborts on receiving flag sequence 0 throttles, 0 enables 0 overruns 0 transmitter underruns 0 transmitter CTS losts 0 aborted short frames R1></pre>	<pre>R2>show controllers serial0/3/0 Interface Serial0/3/0 Hardware is PowerQUICC MPC860 DTE V.35 TX and RX clocks detected idb at 0x81081AC4, driver data structure at 0x81084AC0 SCC Registers: General [GSMR]=0x2:0x00000000, Protocol-specific [PSMR]=0x8 Events [SCCE]=0x0000, Mask [SCCM]=0x0000, Status [SCCS]=0x00 Transmit on Demand [TODR]=0x0, Data Sync [DSR]=0x7E7E Interrupt Registers: Config [CICR]=0x00367F80, Pending [CIPR]=0x0000C000 Mask [CIMR]=0x00200000, In-srv [CISR]=0x00000000 Command register [CR]=0x580 Port A [PADIR]=0x1030, [PAPAR]=0xFFFF [PAODR]=0x0010, [PADAT]=0xCBFF Port B [PBDIR]=0x09C0F, [PBPAPAR]=0x0800E [PBODR]=0x00000, [PBDAT]=0x3FFFD Port C [PCDIR]=0x00C, [PCPAR]=0x200 [PCSO]=0xC20, [PCDAT]=0xDF2, [PCINT]=0x00F Receive Ring rmd(68012830): status 9000 length 60C address 3B6DAC4 rmd(68012838): status B000 length 60C address 3B6D444 Transmit Ring tmd(680128B0): status 0 length 0 address 0 tmd(680128B8): status 0 length 0 address 0 tmd(680128C0): status 0 length 0 address 0 tmd(680128C8): status 0 length 0 address 0 tmd(680128D0): status 0 length 0 address 0 tmd(680128D8): status 0 length 0 address 0 tmd(680128E0): status 0 length 0 address 0 tmd(680128E8): status 0 length 0 address 0 tmd(680128F0): status 0 length 0 address 0 tmd(680128F8): status 0 length 0 address 0 tmd(68012900): status 0 length 0 address 0 tmd(68012908): status 0 length 0 address 0 tmd(68012910): status 0 length 0 address 0 tmd(68012918): status 0 length 0 address 0 tmd(68012920): status 0 length 0 address 0 tmd(68012928): status 2000 length 0 address 0 tx_limited=1(2) SCC GENERAL PARAMETER RAM (at 0x68013C00) Rx BD Base [RBASE]=0x2830, Fn Code [RFCR]=0x18 Tx BD Base [TBASE]=0x28B0, Fn Code [TFCR]=0x18 Max Rx Buff Len [MRBLR]=1548 Rx State [RSTATE]=0x0, BD Ptr [RBPTR]=0x2830 Tx State [TSTATE]=0x4000, BD Ptr [TBPTR]=0x28B0 SCC HDLC PARAMETER RAM (at 0x68013C38) CRC Preset [C_PRES]=0xFFFF, Mask [C_MASK]=0xF0B8 Errors: CRC [CRCEC]=0, Aborts [ABTSC]=0, Discards [DISFC]=0 Nonmatch Addr Cntr [NMARC]=0 Retry Count [RETRC]=0 Max Frame Length [MFLR]=1608 Rx Int Threshold [RFTHR]=0, Frame Cnt [RFCNT]=0 User-defined Address 0000/0000/0000/0000 User-defined Address Mask 0x0000 buffer size 1524 PowerQUICC SCC specific errors: 0 input aborts on receiving flag sequence 0 throttles, 0 enables 0 overruns 0 transmitter underruns 0 transmitter CTS losts 0 aborted short frames R2></pre>

Results of ping tests between all PCs :

From PC1:

```
C:\>ping 192.168.10.1
```

Pinging 192.168.10.1 with 32 bytes of data:

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=5ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
```

Ping statistics for 192.168.10.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms

```
C:\>ping 192.168.20.10
```

Pinging 192.168.20.10 with 32 bytes of data:

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
```

Ping statistics for 192.168.20.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

```
C:\>ping 192.168.30.10
```

Pinging 192.168.30.10 with 32 bytes of data:

```
Reply from 192.168.30.10: bytes=32 time=37ms TTL=126
Reply from 192.168.30.10: bytes=32 time=2ms TTL=126
Reply from 192.168.30.10: bytes=32 time=1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=2ms TTL=126
```

Ping statistics for 192.168.30.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 37ms, Average = 10ms

```
C:\>tracert 192.168.30.10
```

Tracing route to 192.168.30.10 over a maximum of 30 hops:

```
  0 ms  1 ms  7 ms 192.168.10.1
  0 ms  1 ms  0 ms 10.0.0.2
  0 ms  0 ms  1 ms 192.168.30.10
Trace complete.
```

From PC2:

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time=1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=2ms TTL=126
Reply from 192.168.30.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

From PC3:

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 5ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=2ms TTL=126
Reply from 192.168.10.10: bytes=32 time=10ms TTL=126
Reply from 192.168.10.10: bytes=32 time=20ms TTL=126
Reply from 192.168.10.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 20ms, Average = 8ms
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	PC0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC2	PC1	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC2	PC0	ICMP		0.000	N	5	(edit)	(delete)

Screenshot of traceroute between PC1 and PC3 :

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>tracert 192.168.30.10

Tracing route to 192.168.30.10 over a maximum of 30 hops:

  1  0 ms      1 ms      1 ms      192.168.10.1
  2  1 ms      0 ms      1 ms      10.0.0.2
  3 10 ms      1 ms      1 ms      192.168.30.10

Trace complete.

C:\>
```

Answers to validation questions :

1. What is the difference between a trunk link and an access link :

- **Access link:** carries traffic for a single VLAN. It is used between a switch and an end device (PC, printer).
- **Trunk link:** carries multiple VLANs simultaneously using tagging (802.1Q). It is used between switches or between a switch and a router (router-on-a-stick).

2. Why are sub-interfaces used on routers R1 and R2 :

Sub-interfaces are used to allow the router to handle multiple VLANs on a single physical interface (Gig0/0).

This is the principle of **router-on-a-stick**:

- Each sub-interface corresponds to a VLAN
- Each sub-interface has a dot1Q encapsulation
- Each sub-interface has an IP address serving as the default gateway

3. What is the role of the OSPF protocol in this network :

OSPF allows:

- Automatic route exchange between R1 and R2
- Discovery of remote networks
- Dynamic routing between the two sites via the WAN

Without OSPF, the VLAN networks on site 1 could not reach those on site 2 (and vice versa).

4. How many hops are there between PC1 and PC3 :

Logical path:

PC1 → R1 → R2 → PC3

There are 2 routers crossed, so **2 hops**.

5. What happens if the link between C1 and C2 is disconnected :

- Redundancy at the Core level is lost.
- WAN communication between R1 and R2 continues (because it goes through the serial link).
- The VLANs of each site continue to function locally.
- There is no longer a backup link between the two Core switches.

6. Why is the clock rate configured only on R1 for the serial link :

Because **R1 is configured as DCE**:

- The DCE provides the clock signal.
- The DTE receives the clock signal.

Therefore, only the router on the DCE side must have the command:

clock rate 64000

7. Which command is used to verify the routing table on a router :

Command: **show ip route**

It displays:

- Connected routes
- OSPF routes
- Learned remote networks

8. How to verify that OSPF is working correctly between R1 and R2 :

Main commands:

- show ip ospf neighbor
- show ip route
- show ip protocols
- show ip ospf interface brief

If OSPF is working:

- The routers appear as neighbors (Full state)
- Remote routes are visible in the routing table (marked by O)

9. What is the difference between DTE and DCE in a serial connection :

- **DCE (Data Circuit-terminating Equipment):** provides the clock signal.
- **DTE (Data Terminal Equipment):** receives the clock signal.

In this lab:

- R1 = DCE
- R2 = DTE

10. Why do we need to add a HWIC-2T module to the 2911 routers :

The 2911 router does not have serial ports by default.

The HWIC-2T module is necessary to:

- Add Serial interfaces
- Allow WAN link simulation
- Configure DTE/DCE

Without this module, the Serial0/2/0 interface does not exist.